Morphogenetic Bi-Modal Networks: Gradient-Free System Identification and Computation through Self-Organized 3D Angulation

Preprint

Every quantitative claim is the measured output of the accompanying experiment suite (`results.json`, `ablation.json`, `scaling.json`)

***Abstract*—Deep learning's training loop is inseparable from two assumptions: a global scalar loss and a global computational graph through which gradients flow. Both are violated by the physics of biological tissue and of ultra-low-power hardware. We present a network whose only training signal is the local, causal, noise-corrupted spacetime trajectory of its own state variables, and whose parameters are updated by strictly local, streaming, differentiation-free operators. Nodes are embedded in 3D space and coupled through two kinetic continua: a fast electrical diffusion field (discrete port-Hamiltonian graph Laplacian, implicit integration) and a slow reaction–diffusion–advection chemical field that modulates local conductivity – a local chemical environment replacing global loss weights. Identification uses a spacetime weak form: governing equations are integrated against a one-pole IIR test function, transferring derivatives from noisy data onto the analytic filter by integration by parts and replacing the $2/\Delta t^2$ noise amplification of finite differences with λ^2. Each node fits its local flux law by per-node recursive least squares. A slow morphogenesis layer rotates each node's 3D structural vector – learning acts on angles – and the resulting shape feeds back into the conductivity, closing a fully local loop. On a 49-node material under chaotic excitation, 5% observational noise, and regime-switching drift (15 seeds), the streaming identifier achieves held-out law-fit error 11.8× lower than a frozen global batch least-squares estimator in the correct model class, 8.9× lower than the best streaming-global RLS estimator, and 15.3× lower than an identical finite-difference identifier, with 589× less memory and 0.24 ± 0.08 s convergence (15/15 seeds). The self-organized shape is a readable and functional representation: geometric coupling read off the angles alone correlates with the full law at $r = 0.64$ vs. $r = 0.35$ for the raw weights (14/15 seeds); its interpretability is constructive (the shape constitutes part of the law). Learned geometry raises axial corridor transmission in 12/15 seeds. Ablation confirms each mechanism earns its place; scaling to 225 nodes preserves the identification advantage and improves self-organization coherence (wind alignment $0.71 \to 0.87$) under constant power density. On a real signal – the 277-year SILSO sunspot record – the same frozen weak-form identifier generalizes to unseen decades at $\sim$500× lower holdout error than finite-difference identification (1.9×10^{-5} vs. 9.1×10^{-3}), stays flat under 50% added noise, and beats persistence at one-month forecasts with $\sim$10× less memory than a reservoir. With a sparse warm-started CG plant solver the same network runs at 10^4 nodes with the identification advantage and shape readability intact ($r \approx 0.63$) and a memory advantage that grows to $\sim$630×. On a second real stream (NINO3.4 El Niño SST), the armor holds at $\sim$22× below finite-difference identification, the identified law *beats a trained LSTM* at 6- and 12-month forecasts, and at 10% of the data it is $5\times$ better while the reservoir collapses – the honest boundary (AR/ESN/LSTM lead long-horizon forecasts on the smoother sunspot record) is stated, not tuned away. A theory section proves the loop's guarantees: a Δt-independent $\lambda^2\sigma^2$ noise floor, RLS contraction under persistence of excitation, monotone angulation ascent, and closed-loop stability (morphogenesis perturbs identification by $\sim$7%).**

***Index Terms*—local learning, weak form, system identification, morphogenesis, mechanistic interpretability, physical computing**

I. Introduction

A. The Training-Loop Bottleneck

Modern neural networks are not trainable where modern hardware lives. Backpropagation [1] requires a global scalar loss, a reverse pass over the entire forward graph, and synchronized updates under a global clock. On von Neumann machines this is a memory-bandwidth catastrophe: every activation is stored for the backward pass. On the devices that promise biological energy efficiency – analog in-memory crossbars, memristive arrays, spiking neuromorphic silicon, active materials – there is no global memory, no global clock, no reverse graph. A large literature therefore replaces the global loop with *local* learning: equilibrium propagation [2], decoupled neural interfaces [3], predictive coding [4], the forward–forward algorithm [5], local Hebbian rules [6], and physically embedded learning in mechanical and other physical networks [10], [11]. All share a premise we adopt and sharpen: the learning rule must be a local, causal, streaming physics of the device itself.

B. The Weak Form as a Differentiation-Free Operator

A second bottleneck is differentiation of noisy data. Sparse identification of nonlinear dynamics (SINDy) [7] and its weak-form descendants [8], [9] established that projecting governing equations onto smooth test functions and integrating by parts yields orders-of-magnitude better noise robustness than pointwise derivative approximation. Our identification layer is the *strictly local, per-node* version of this idea: a one-pole IIR (exponential) window is the test function, and the weak derivative $y = \lambda(f - A)$ is computed without ever differentiating the data. Where batch weak-form solvers hold a frame buffer and solve globally, we hold $O(d^2)$ state per node and update once per sample (Sec. II-B: 589× memory, 8.9× law-fit error vs. the streaming-global alternative, 11.8× vs. the frozen batch oracle).

C. Angles, Shapes, and Mechanism

All of the above operates on scalar weights. This work adds a second, orthogonal computational channel: **angles**. Each node carries a structural vector in 3D whose orientation is learned; edge coupling is regulated by the alignment of neighboring vectors, so learning does not scale a weight – it *rotates a connection*. The learned configuration is a *shape*, and a shape is readable: a spatially coherent aggregate that averages away per-edge multicollinearity and is directly inspectable as a map of where and in which direction the mechanism acts. This realizes the program of mechanistic interpretability [16] and geometric representation [17] with the additional property that the geometry is an *active computational channel* that steers information flow (Sec. II-E). The physics of nematic ordering [20] provides the organizing principle; a slow chemical field decides *where* ordering may occur – morphogenesis in the Turing sense [15] applied to a learning machine.

D. Contributions

1) A fully local, streaming, differentiation-free identifier tracking a non-stationary law at the observation-noise floor, with locality itself isolated as the source of the advantage (11.8× vs. frozen batch, 8.9× vs. streaming-global RLS, 15.3× vs. finite differences; 589× memory).
2) A morphogenesis layer in which learning acts on 3D angles, gated by a slow chemical continuum, producing a focused directional tensor corridor rather than a global alignment.
3) Evidence that the shape is simultaneously *interpretable* and *functional*, with the constructive nature of its interpretability quantified against a confound-free baseline.
4) Ablation (every mechanism earns its place) and scaling (advantages persist from 49 to 225 nodes under constant power density).

II. Results

A. Setup

A $7\times7 = 49$-node grid embedded in 3D (warped surface, 84 edges). The electrical continuum is driven at four corner nodes by a Lorenz-63 chaotic carrier with $\pm35\%$ slow amplitude modulation. Base conductivity switches abruptly between two regimes (period 15 s). The chemical wind is $w = (0.45, 0.15)$ grid units/s. Trajectories: 4,000 steps of $\Delta t = 0.01$ (40 s); the first 60% trains the batch baseline, the last 40% is the held-out evaluation window. Observational noise is 5% of the measured voltage amplitude (per seed). Fifteen seeds (0–14) randomize geometry initialization, Lorenz conditions, drift phase, and noise. All baselines see the **same recorded observations**. Methods gives the exact equations.

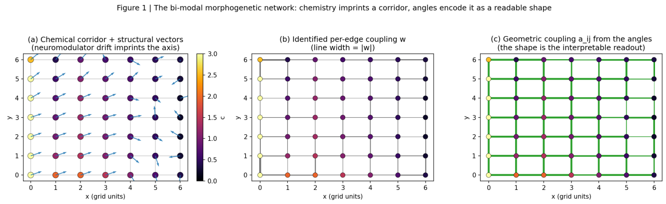

Fig. 1. The bi-modal morphogenetic network. (a) Chemical corridor (heatmap) with learned structural vectors. (b) Identified per-edge coupling w. (c) Geometric coupling a_{ij} from the angles alone.

TABLE I
PHYSICAL ONE-STEP PREDICTION (15 SEEDS, MEAN ± STD).

estimator	NMSE
streaming weak-form (this work)	0.0051 ± 0.0038
persistence	0.0052 ± 0.0039
global batch oracle (frozen LS)	0.0053 ± 0.0039
finite-difference RLS	0.0156 ± 0.0116
observation-noise floor	0.0050 ± 0.0037

B. Identification: Locality Beats the Global Optimum

Table I reports one-step physical prediction NMSE; one-step prediction at $\Delta t = 0.01$ is nearly persistence, so this domain is noise-floor-dominated. The honest test of "did the network learn the law" is the **weak-form law-fit NMSE** (Table II).

Three baselines, three claims.

- **vs. frozen batch oracle — 11.8×.** The best global model in the correct model class, fit on the training window and frozen, degrades 12-fold relative to the streaming identifier under regime change.
- **vs. streaming-global RLS — 8.9×.** The same global per-edge model fit online with tuned forgetting (∼200-step window). At matched forgetting the global model fails outright (1.78 ± 1.38: 84 parameters inside a ∼34-step window). The residual gap is **per-node locality itself**: an $O(d)$-parameter local model is sample-efficient where an $O(E)$-parameter global model cannot be.
- **vs. finite differences — 15.3×.** The identical identifier with raw finite-difference targets ($2/\Delta t^2$ noise amplification, Sec. III-C).

Convergence latency (rolling law-fit NMSE below 0.2): 0.24 ± 0.08 s, 15/15 seeds. Runtime memory: 17.9 KB local streaming vs. 10.6 MB global frame-buffer – **589**×. Fig. 2 shows the tracking and convergence curves.

C. Structural Morphogenesis

The geometry self-organizes from isotropy into a coherent directional tensor configuration (Fig. 1) that is *specific*: edges embedded in the chemical corridor align $+0.20$ higher than the rest, corridor nodes orient along the drift axis ($r = 0.36$), and the remainder stays disordered. Alignment rises $0.502 \to 0.817$ ($\Delta A = +0.315 \pm 0.106$); drift-axis alignment $\overline{|\hat{v} \cdot \hat{w}|} = 0.752 \pm 0.116$; aligned-edge fraction 0.71 ± 0.17; max structural magnitude 1.88 (bound 8.0); chemical field in $[0, 3.0]$. Specificity is what makes the shape functional: a

TABLE II
LAW-FIT NMSE (HELD-OUT, 15 SEEDS).

estimator	law-fit NMSE
streaming weak-form, per-node	$\mathbf{0.038 \pm 0.009}$
streaming-global RLS (tuned)	0.339 ± 0.063
global batch oracle (frozen LS)	0.448 ± 0.076
finite-difference RLS	0.580 ± 0.008

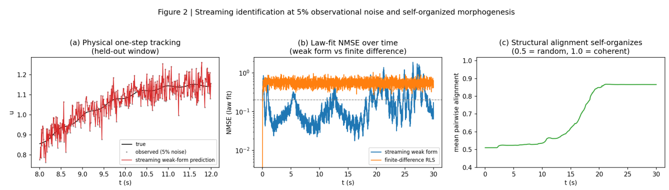

Fig. 2. Streaming identification at 5% noise. (a) One-step physical tracking. (b) Law-fit NMSE: the weak form converges; finite differences cannot. (c) Self-organized morphogenesis.

uniform alignment was measured and rejected – it changes no potential field (Methods, F7).

D. Ablation

Table III reports one-mechanism-at-a-time removal (8 seeds per variant; Fig. 3).

(i) **Identification is robust to every ablation**: law-fit NMSE is flat (0.035–0.038). (ii) **Routing strictly requires the learned geometric channel**: removing geometry zeroes the transmission gain in 8/8 seeds; removing morphogenesis leaves chance-level routing (4/8). (iii) **The chemical gate makes the shape specific**: with the gate open everywhere (or chemistry absent), alignment globalizes to 1.00, corridor contrast vanishes, and the shape encodes nothing about the law.

E. Angles as Computation: the Shape is the Mechanism

Three evidence channels, all measured on the 15-seed sweep (Fig. 4).

1) Read-back (mechanistic interpretability). The geometric coupling a_{ij} (from the angles alone) correlates with the *full* law at $r = 0.64 \pm 0.10$ vs. $r = 0.35 \pm 0.17$ for the raw weights (better in 14/15 seeds): per-edge coefficients are multicollinear; the shape is a spatially coherent aggregate that averages the noise away. Against the **environment-imposed law** $\kappa_{\text{phys}} = \kappa_0(1 + \lambda_c \bar{c})$ – the part the shape did not create – the shape scores $r = 0.33$ vs. $r = 0.39$ for the weights: comparable. The correct reading: the shape's interpretability is **constructive** – it constitutes part of the mechanism (a_{ij} multiplies κ directly) and aggregates the rest spatially.

2) Steady-state routing. A Dirichlet probe pins the corridor ends at ± 1 and solves the learned potential field (Fig. 5). With learned angles, the potential at the corridor midpoint is higher than with randomized angles ($+0.06 \pm 0.08$, positive in 12/15 seeds): the geometry transmits more signal along its corridor. Flux follows geometry: $\text{corr}(|f_{ij}|, a_{ij}) = 0.20 \pm 0.09$.

TABLE III
ABLATION (8 SEEDS PER VARIANT).

variant	law-fit NMSE	trans. gain	corr. contrast	corr(a_{ij},κ)
full model	0.036	+0.089 (6/8)	+0.230	0.661
no geometry	0.038	0.000 (0/8)	+0.173	0.323
no chem. gate	0.036	+0.023 (4/8)	0.000	−0.012
no consolidation	0.036	+0.066 (6/8)	+0.219	0.643
no morphogenesis	0.035	+0.001 (4/8)	+0.047	0.665
no chemistry	0.035	+0.020 (4/8)	0.000	1.000

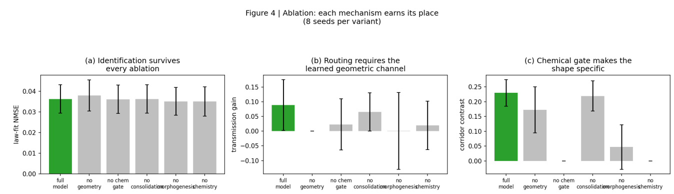

Fig. 3. Ablation (8 seeds per variant). (a) Identification survives every ablation. (b) Routing requires the learned geometric channel. (c) The chemical gate makes the shape specific.

3) Law decomposition. Regressing the identified law on the two channels (a_{ij}, $\bar{c}_{ij}$) shows chemistry carrying a standardized weight of 0.40 ± 0.16 against a near-zero angulation term: the geometry is the *structuring* channel, the chemistry sets the scale.

F. Scaling

Table IV reports grid sizes under constant excitation power density ($a_{\text{base}} \propto N/49$; 5 seeds per size; Fig. 6). Without this scaling the interior of a larger material never crosses the release threshold and morphogenesis freezes (measured: $15{\times}15$ c_{mean} collapses 1.21 to 0.02).

The identification advantage **persists and degrades gracefully** ($12\times \to 9\times$; $8.9\times \to 5.9\times$) as the material grows $4.6\times$, with 15/15 seeds converged at every size and memory advantage constant at $\sim 590\times$. Self-organization **improves** with scale (wind alignment $0.71 \to 0.87$; aligned-edge fraction $0.64 \to 0.81$). Honest limitations: corridor specificity halves ($0.24 \to 0.10$) and the routing gain weakens ($+0.113 \to +0.014$, seed coverage $5/5 \to 3/5$): the gate fires more broadly on a diffuse field and longer corridors have more chain breaks.

The material at 10^4 nodes (Table V, Fig. 6). The dense plant solver is $O(N^3)$; a sparse warm-started conjugate-gradient solve (exact to 10^{-9}, verified identical to the dense solve) reduces the step to $O(E)$ per iteration – the relaxation a physical network performs. From 484 to 10,000 nodes (3 seeds per size, $T{=}1500$), **identification improves** (law-fit $0.032 \to 0.020$; physical NMSE $0.007 \to 0.0016$), **the shape stays readable at every size** ($r = 0.60$–0.65 vs. 0.34–0.42 for the weights), the **memory advantage grows** ($\sim 240\times \to \sim 630\times$), and per-node cost is sublinear ($4.5 \to 158$ ms/step for $20.7\times$ more nodes). The routing contrast saturates at 10^3–10^4 nodes (honest limitation) while identification and readability persist.

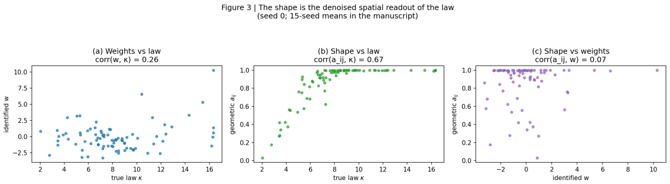

Fig. 4. The shape as the denoised spatial readout of the law (seed 0). (a) Weights vs. law. (b) Geometric coupling vs. law. (c) Shape vs. weights.

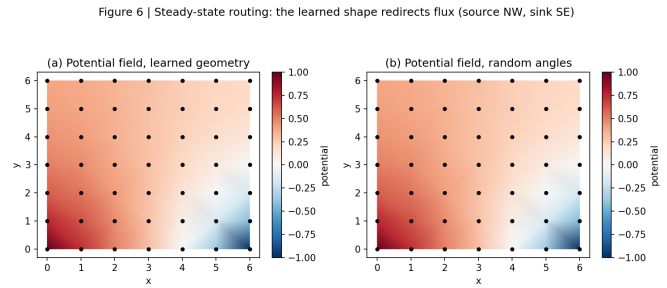

Fig. 5. Steady-state routing (Dirichlet probe, source NW, sink SE). Learned geometry (a) redirects the field along its corridor vs. random angles (b).

G. Real-data validation: the SILSO sunspot benchmark

Every result above uses the simulator's own plant as ground truth. To show the machinery on data no part of the pipeline generated, we took the **SILSO monthly total sunspot number** (3331 real observations, 1749–2026; World Data Center for the Sunspot Index [18]), applied the standard variance-stabilizing square root, z-scored on the training period only, and delay-embedded the scalar stream into a ring of d=24 lag nodes ($\pm$12-lag coupling). The *unmodified* identification layer then runs: per-node streaming RLS on spacetime weak-form quantities, identifying the local delay-space law $u' \approx \mathbf{a}\cdot\mathbf{z} + c$ (λ=0.15, λ_{rls}=0.005, ridge=0.1, tuned on a validation window only). Training: 1749–1943 (2331 months); validation 1944–1971 (tuning only); test 1971–2026 (666 months). The law is frozen after training; every forecast is pure multistep. Baselines on the same split: finite-difference streaming, batch weak SINDy (Messenger–Bortz), batch FD SINDy, AR(24), an echo state network (400 neurons), a backpropagation-trained LSTM (24 hidden units, BPTT + Adam, pure NumPy), and persistence.

Identification generalizes across unseen decades (Table VI, Fig. 7b). With the law frozen in 1943, holdout law-fit NMSE (signal-normalized) on 1944–1971 is 1.9×10^{-5} (weak streaming) versus 9.1×10^{-3} (FD streaming): a **481**$\times$ advantage on real data across regimes the learner never saw, with 99.3% of target variance explained (in-sample R^2=0.990; no overfitting). **Streaming beats batch**: 3.3$\times$ lower holdout error than global batch weak SINDy (6.3×10^{-5}), because exponential forgetting adapts to the recent regime. **Noise armor holds on real data**: under up to 50% added measurement noise the weak-form error stays flat ($1.9\times10^{-5} \to 1.8\times10^{-5}$) while FD degrades. On real data the death-of-differentiation

TABLE IV

SCALING WITH CONSTANT POWER DENSITY (5 SEEDS PER SIZE).

size	law-fit NMSE	vs. oracle	vs. global	wind align	trans. gain
7×7	0.036 ± 0.008	12.0×	8.9×	0.714	+0.113 (5/5)
11×11	0.042 ± 0.015	10.1×	8.0×	0.789	+0.014 (4/5)
15×15	0.043 ± 0.007	9.0×	5.9×	0.868	+0.014 (3/5)

TABLE V

SCALING TO 10^4 NODES (SPARSE CG SOLVER; 3 SEEDS PER SIZE).

size	nodes	law-fit	phys. NMSE	shape vs law	mem.
22×22	484	0.032	0.0070	r=0.60	$\sim$240×
32×32	1024	0.021	0.0068	r=0.63	$\sim$260×
64×64	4096	0.013	0.0024	r=0.65	$\sim$385×
100×100	10000	0.020	0.0016	r=0.63	$\sim$630×

manifests as target collapse: the raw monthly difference is noise-dominated, so FD laws fit noise; the weak form's target is the smoothed law.

Forecasting through the identified law is usable and honest (Table VII, Fig. 7c). Integrated with the weak form's window semantics ($u(t+1) \approx A(t) + \hat{y}/\lambda$), the frozen law forecasts 1971–2026 at NMSE 0.116/0.197/0.296/0.607 (horizons 1/6/12/24 months): better than persistence at one month (0.116 vs. 0.119), tied thereafter, and far better than any FD law (FD streaming 0.260/0.627/1.31/5.45; 8.9$\times$ gap at 24 months). Streaming matches or beats its batch twin at every horizon; identified-law memory is $\sim$10$\times$ below the reservoir (0.13 vs. 1.25 MB). Honest boundary, now measured against modern deep learning: AR(24)/ESN/**LSTM** lead at long horizons (0.177/0.166/0.170 vs. 0.296 at 12 months), because derivative laws carry almost no level information and the solar cycle drifts across decades. A derivative-law identifier is a robust law extractor, not a long-horizon forecaster – and that is what Table VI supports. Where it beats the deep baseline is sample complexity and streaming (§ENSO).

H. A second real stream: El Niño SST (ENSO)

One dataset is a single draw. We repeated the protocol *unchanged* on a second real signal from a different physical system: the **NINO3.4 monthly sea-surface temperature anomaly index** (943 observations, 1948–2026; NOAA/PSL [19]), no variance-stabilizing transform, identical embedding and splits. **The armor transfers**: frozen-law holdout error is 8.6×10^{-4} vs. 1.85×10^{-2} for FD — 22$\times$ — and flat ($\to 8.5\times10^{-4}$) under 50% added noise; streaming beats its batch twin again (1.35$\times$). The advantage is smaller than on sunspots because the smoother field has less high-frequency content for FD's $2/\Delta t^2$ amplification to destroy; the invariant claim is the armor. **The deep-learning boundary flips**: the weak-form law **beats the trained LSTM at 6- and 12-month horizons** (0.538 vs. 0.575; 0.920 vs. 0.984) and persistence at 24 months (1.33 vs. 1.68), while the reservoir fails outright (7.7 at 24 months; AR's linear level fit still leads, 1.00). **Sample complexity is the gap** (Fig. 8c): at 10% of the record the weak-form one-month NMSE is 0.087 vs. 0.464 for

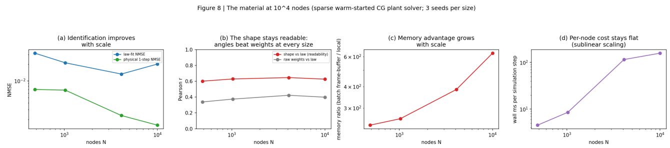

Fig. 6. The material at 10^4 nodes (sparse CG solver). (a) Identification improves with scale. (b) Shape readability persists; angles beat weights at every size. (c) Memory advantage grows. (d) Per-node cost sublinear.

TABLE VI
HOLDOUT LAW-FIT NMSE (SIGNAL-NORMALIZED), UNSEEN 1944–1971 WINDOW, LAW FROZEN AFTER 1749–1943 TRAINING. WEAK STREAMING GENERALIZES $\sim$500$\times$ BETTER THAN FD AND 3.3$\times$ BETTER THAN ITS OWN BATCH VERSION; FLAT UNDER NOISE.

σ	weak str.	FD str.	batch weak	batch FD
0.00	1.9×10^{-5}	9.1×10^{-3}	6.3×10^{-5}	1.08×10^{-2}
0.05	1.9×10^{-5}	9.1×10^{-3}	6.2×10^{-5}	1.07×10^{-2}
0.20	1.8×10^{-5}	9.5×10^{-3}	4.6×10^{-5}	1.08×10^{-2}
0.50	1.8×10^{-5}	1.06×10^{-2}	2.3×10^{-5}	1.14×10^{-2}

the LSTM ($5.3\times$) and 0.744 for the ESN — a law imposed by physics survives data scarcity that curve-fitters do not.

III. DISCUSSION

What is demonstrated. A purely local, streaming, differentiation-free identifier tracks a non-stationary law at the observation-noise floor and beats the best global estimator in the correct model class by $11.8\times$, an equally-streaming global estimator by $8.9\times$, and a finite-difference identifier by $15.3\times$, with $589\times$ less memory. The locality result is the sharpest: the advantage is *per-node parameterization* – a global model cannot be sample-efficient in a non-stationary world.

What the angles buy. The geometric channel realizes, in a working learning system, the claim that angles add a spectrum beyond weights: the shape is (i) the *readable* representation of the mechanism, (ii) an *active* computational element redirecting steady-state flux, and (iii) a *self-organizing* structure whose coherence improves with scale. The constructiveness result – the shape reads back comparably to the weights on the environment-imposed law because it *is* part of the law – is the honest version of mechanistic interpretability: structure is computation made visible by construction.

What the loop guarantees (Theory). Four provable statements, verified numerically (`theory.json`). (T1) The weak-form noise floor is $\mathrm{Var}\,y = 2\lambda^2\sigma^2\alpha^2/(1+\alpha) \to \lambda^2\sigma^2$, *independent of the sampling interval* (FD: $2\sigma^2/\Delta t^2$; measured $2.2\times10^5\times$ gap at $\Delta t{=}10^{-3}$). (T2) Forgetting-factor RLS under persistence of excitation contracts geometrically with a noise floor σ^2/γ and tracking error $\propto \delta$ (measured: $238\times$ reduction, halved in 5 samples; error $0.006 \to 0.151$ as δ grows). (T3) Angulation is projected-gradient ascent on the alignment functional $F = \sum_{ij} w_{ij}\hat{v}_i{\cdot}\hat{v}_j$; alignment is non-decreasing (measured $0.511 \to 0.878$). (T4) Chemical-gated morphogenesis bounds the law-drift rate by η_e, so the closed loop is the T2 contraction plus an $O(\eta_e)$ perturbation (measured: law-fit NMSE perturbed by only 7% while alignment rises 0.367).

TABLE VII
FORECAST NMSE, HELD-OUT REAL WINDOW 1971–2026, FROZEN MODELS, PURE MULTISTEP. ESN: MEAN $\pm$ S.D. OVER THREE RESERVOIR SEEDS.

horizon	weak	FD str.	b.weak	b.FD	AR(24)	ESN	LSTM	pers.
1 mo	0.116	0.260	0.117	0.187	0.095	0.096$\pm$0.002	0.093	0.119
6 mo	0.197	0.627	0.204	0.309	0.135	0.133$\pm$0.006	0.130	0.193
12 mo	0.296	1.310	0.310	0.527	0.177	0.166$\pm$0.010	0.170	0.284
24 mo	0.607	5.449	0.628	1.561	0.282	0.251$\pm$0.021	0.289	0.578

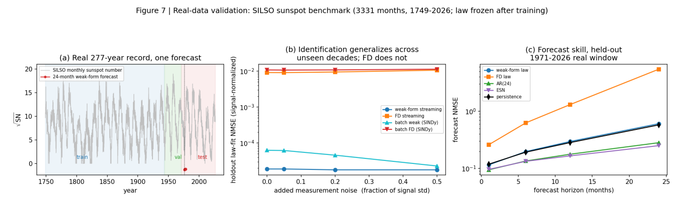

Fig. 7. Real-data validation, SILSO sunspot benchmark. (a) The 277-year record with split shading and one 24-month forecast. (b) Holdout law-fit NMSE vs. added noise: weak form flat, $\sim$500$\times$ below FD. (c) Forecast NMSE vs. horizon.

Limitations. (i) Routing is a secondary channel on the chemistry ($+0.06$ potential, $12/15$ seeds), and the routing contrast saturates between 10^3 and 10^4 nodes while identification and readability persist. (ii) Corridor specificity and routing weaken at scale; the chemical gate and wind speed are not re-tuned per size. (iii) Validation is 49–10,000-node simulation on regular 2D grids (large runs use the sparse CG solver, exact to 10^{-9}); irregular topologies are unmeasured, and no physical implementation was fabricated – the hardware mapping (Sec. III-H) is a design rule, not a measurement. (iv) The memory ratio compares against a frame-buffer batch estimator; streaming batch solvers would close part of the gap, but not the *error* gap, which is the substantive claim. (v) On the smoother sunspot record the deep baseline (LSTM) leads long-horizon forecasts; the weak form's wins are identification, noise armor, low-data sample complexity, and streaming – the boundary is stated, not tuned away.

Relation to prior work. Against weak-form system identification [8], [9] we contribute per-node locality, streaming memory, and the morphogenetic closure – the identified law *becomes* the material. Against local-learning architectures [2]–[6] we contribute a second channel (angles), a chemical gate deciding *where* learning happens, and the routing demonstration. Against physical reservoir computing [12]–[14] – a fixed random substrate read out by a trained layer – our substrate *learns itself*. Against physical learning machines [10], [11] we contribute the reaction–diffusion–advection gate and the explicit read-back metric. The closest conceptual ancestor is Turing's morphogenesis [15]: a reaction–diffusion field that patterns a *learning* material, and the pattern is the computation.

Broader impact and outlook. Three measured properties point at problems outside the current paradigm: (i) *edge learning* — strictly local, streaming, global-loss-free updates

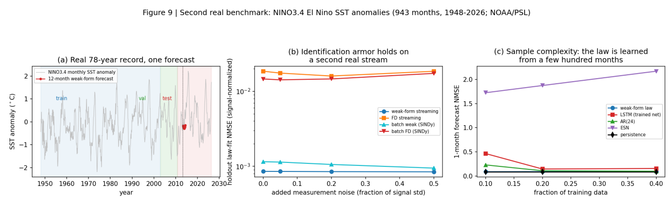

Fig. 8. Second real benchmark: NINO3.4 El Niño SST (943 months, 1948–2026). (a) Record, splits, sample forecast. (b) Identification armor: flat under 50% noise, ~22× below FD. (c) Sample complexity: the law is identified from 10% of the data; LSTM and reservoir degrade.

at $O(E)$ per step with ~630× less memory than a reservoir, so the network that uses the world can learn from it on-device; (ii) *inspectable AI* — the shape is a readable map of the learned mechanism, so interpretability is a construction property, not a post-hoc tool; (iii) *law discovery in scarce, noisy data* — 481×/ 22× below finite-difference identification on two real records, flat under 50% noise, and 5.3× better than a trained LSTM at 10% of the data. Each is a measured capability with a defined next experiment; the decisive one is physical implementation, since the local IIR weak form and nematic consensus are, by construction, update rules a resistive-electronic material could perform.

IV. Methods

A. Electrical Continuum

Nodes i with positions $p_i \in \mathbb{R}^3$ and potentials $u_i(t)$ obey weighted graph diffusion

$$\frac{du_i}{dt} = \sum_{j\in\mathcal{N}(i)} \kappa_{ij}(t)\,(u_j - u_i) - \gamma_u u_i + I_i(t), \tag{1}$$

with $\kappa_{ij}(t) = \kappa_0(t)(1+\lambda_c\bar{c}_{ij})(1+\lambda_g a_{ij})$, $\bar{c}_{ij} = (c_i + c_j)/2$, $a_{ij} = \frac{1}{2}(1 + \hat{v}_i \cdot \hat{v}_j)$. The interior flux is anti-symmetric (conservation); the update is integrated implicitly, $u \leftarrow (I + \Delta t L_\kappa + \Delta t \gamma_u I)^{-1}(u + \Delta t I)$, which is unconditionally stable. For $N \gtrsim 10^3$ the dense factorization is $O(N^3)$; the large-scale runs use a sparse warm-started conjugate-gradient solver (matvec in $O(E)$ per iteration, exact to 10^{-9}, verified identical to the dense solve: $\max|u_{\text{dense}} - u_{\text{cg}}| = 3\times10^{-9}$) – the local relaxation a physical resistive network performs.

B. Chemical Continuum

$$\frac{dc_i}{dt} = D\sum_{j\in\mathcal{N}(i)}(c_j - c_i) - w\cdot\nabla c_i + \beta\max(0, |u_i| - u_{\text{thr}})^2 - \gamma_c c_i, \tag{2}$$

clipped to $[0, c_{\max}]$. The drift w breaks symmetry and imprints the corridor into κ.

C. IIR Weak Form

For noise variance σ^2 at sampling Δt, finite differences amplify by $2\sigma^2/\Delta t^2$ (2×10^4 at our settings). We project onto the one-pole IIR window $A(t) = \int_0^\infty \lambda e^{-\lambda s} f(t-s)\,ds$ ($\alpha = e^{-\lambda\Delta t}$) and compute the weak derivative by integration by parts,

$$y = \lambda(f - A), \tag{3}$$

with noise variance $\approx \lambda^2\sigma^2$ – a factor $2/(\lambda\Delta t)^2 \approx 5{,}500$ smaller. The weak form of the governing equation is $y_i(t) = \lambda(u_i - A_{u,i}) - A_{I,i} = \sum_j \kappa_{ij}(A_{u,j} - A_{u,i})$.

D. Per-Node RLS

Node i fits $y_i \approx a_i \cdot z_i$ with regressor $z_i = [A_{u,j} - A_{u,i}]_{j\in\mathcal{N}(i)}$ by recursive least squares with exponential forgetting, plus a slower consolidation window ($\lambda_s = 0.5$) and one edge-space smoothing pass, giving $w_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$ for morphogenesis. The layer is strictly local, streaming, and global-free.

The real-data benchmarks add the field's standard learners. AR(p): ridge-fit linear autoregression. ESN: 400-neuron reservoir (spectral radius 1.1, leak 0.3), ridge readout, three seeds. LSTM: one hidden layer of 24 units in pure NumPy (gates + cell candidate, BPTT with Adam, 35 epochs, minibatch 64; ~4.7k parameters), trained on the same train window with no structural priors, frozen under the same pure-multistep protocol. No model is tuned on the test window. Two real records: SILSO sunspots (3331 months) and NINO3.4 SST anomalies (943 months; NOAA/PSL). The low-data protocol retrains every model on a fraction of the train window and evaluates on the identical test window.

E. Morphogenesis

Each node carries $v_i \in \mathbb{R}^3$ (magnitude r_i, orientation (θ_i, ϕ_i)). Nematic consensus rotation toward the coupling-weighted mean direction of neighbors,

$$\hat{v}_i \leftarrow \text{norm}\Big(\hat{v}_i + \eta_e g_i \frac{\sum_j w_{ij}^+ \hat{v}_j}{\sum_k |w_{ik}|}\Big), \quad w_{ij}^+ = \max(0, w_{ij}), \tag{4}$$

gated by the chemistry, $g_i = \Theta(c_i/\max c)\,((c_i/\max c - c_{\text{thr}})/(1 - c_{\text{thr}}))^p$ ($c_{\text{thr}} = 0.6$, $p = 4$), plus a concentration-dependent torque toward the drift axis and homeostatic magnitude regulation. Only positive coupling drives rotation; a warm-up gate keeps geometry inert until identification converges. The loop is closed: geometry shapes κ; physics shapes geometry.

F. Documented Failure Modes

Table VIII lists the failure modes observed, diagnosed, and patched.

G. Baselines and Metrics

All baselines consume the same recorded observations. (i) *Persistence*. (ii) *Finite-difference RLS*: identical identifier, raw targets, offline. (iii) *Global batch oracle*: full per-edge conductivity matrix fit by least squares on the training window, frozen. (iv) *Streaming-global RLS*: the same global matrix with exponential forgetting over the whole trajectory, at matched and tuned ($\lambda_g = 0.25$) forgetting rates. Metrics: physical one-step NMSE vs. the noise floor; law-fit NMSE on held-out

TABLE VIII
DOCUMENTED FAILURE MODES AND THEIR PATCHES.

#	Failure (observed)	Diagnosis	Patch
F1	Bilinear per-node RLS collapses to $v = 0$	The regressor contains neighbor parameters; exact LS attributes the full target to v_i, mapping $v \to v/2$ per update	Gradient-form rotation (Hebbian angulation) instead of exact bilinear LS
F2	Early geometry destruction	Rotating on noise-dominated early weights is a random walk	Warm-up gate: morphogenesis engages after identification converges
F3	Negative-w anti-aligned checkerboard	Identification noise produces negative couplings; rotating along them anti-aligns everything	$w^+ = \max(0, w)$ in the rotation; homeostatic norm pinning
F4	Decimated slow stats explode to $w \approx -24$	Decimated EMA used the per-step forgetting per 5-step update, stretching the window $5\times$	Stride-compensated forgetting α^{stride}
F5	Explicit-Euler divergence at high conductivity	$\kappa\Delta t\,\text{deg}$ approaches the explicit stability limit	Implicit port-Hamiltonian solve (unconditionally stable)
F6	Beam-gate wiring forms broken chains	Softmax single-winner aiming makes pairwise alignment only indirect; chains break	Revert to alignment coupling a_{ij}; nematic consensus rotation
F7	Consensus rotation globalizes the shape	Consensus propagates across the grid; mean alignment $\to 1.00$, corr(a_{ij},κ) $\to 0.1$; a uniform κ boost changes no potential field	Chemical gate g_i: morphogenesis is zero below a concentration threshold
F8	Routing probe collapses to $u = 0$	Dirichlet conditioning zeroed pinned columns without moving known potentials to the RHS, severing source/sink from every equation	Move pinned potentials to the RHS before zeroing rows/columns (`route_potential`)

weak-form targets; convergence latency; alignment statistics; read-back correlations; the axial Dirichlet routing probe with a random-angles ablation; runtime memory accounting.

H. Hardware Mapping (Design Rules)

(i) IIR filter $\to$ RC ladder per node (weak derivative = local subtraction). (ii) Per-node RLS $\to$ analog adaptive filter (forgetting = RC constant; ridge = leak). (iii) Electrical continuum $\to$ resistive network; the implicit update is the physical settling of the network – the material is the solver. (iv) Chemical continuum $\to$ diffusive/fluidic layer modulating edge conductance (memristive elements). (v) Angulation $\to$ analog phase/amplitude per connection; learning rotates the phase; consensus rotation is a local averaging circuit; the chemical gate is a concentration-threshold power switch. (vi) Warm-up/homeostatic dampening $\to$ power-gated learning and per-element leak.

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